

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Lecture 10 Scribe: Randomized Min-Cut Algorithm

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1 Outline of the Lecture

The lecture covers the following topics:

- Min-Cut Problem (Definition and Applications)
- Max-Flow Min-Cut Theorem
- Deterministic Min-Cut Algorithm (Stoer–Wagner)
- Randomized Min-Cut Algorithm (Karger’s Algorithm)
- Comparison: Deterministic vs Randomized Approaches
- Python Simulation

2 Min-Cut Problem

2.1 Why Use Min-Cut?

We use the min-cut algorithm in various applications to solve problems related to network connectivity, reliability, and optimization.

1. Network Design

Min-cut helps in improving the efficiency of communication and optimizing network flow.

- **Objective:** Improve communication efficiency.
- **Objective:** Optimize network flow.
- **Method:** Identify the minimum capacity cut.

2. Communication Networks

Minimum cut is useful for understanding the vulnerability of networks to failures.

- Evaluates network vulnerability.
- Assists in building robust and fault-tolerant communication networks.

3. VLSI Design

In Very Large Scale Integration (VLSI) design, the algorithm is used for partitioning circuits into smaller components.

- Leads to reduced interconnectivity complexity between components.

3 What is Min-Cut?

3.1 Definition: Cut-Set

A **cut-set** in a graph is a set of edges whose removal breaks the graph into two or more connected components.

3.2 Definition: Minimum Cut

Given a graph $G = (V, E)$ where V is the set of vertices and E is the set of edges. The **minimum cut (min-cut)** problem is to find a minimum cardinality cut-set in G .

- Among all possible cut-sets, select the one with the smallest number of edges.

3.3 Sensitivity of Randomized Algorithms

Min-cut algorithms (like Karger's) are random and sensitive to the initial choice of edges. If the algorithm contracts critical edges early in the process, it might find a smaller cut (or fail to find the true min-cut).

4 Edge Contraction

The main operation in the randomized algorithm is **edge contraction**. Let the edge be (u, v) . Contraction consists of:

1. Merging vertices u and v into one vertex.
2. Eliminating all edges connecting u and v (self-loops).
3. Retaining all other edges (resulting in potential parallel edges).

5 Successful vs. Unsuccessful Runs

- **Successful Min-Cut Run:** An execution of the algorithm where it correctly identifies the minimum cut of the graph.
- **Unsuccessful Min-Cut Run:** An iteration where the algorithm fails to correctly identify the minimum cut (e.g., due to contracting an edge that belongs to the min-cut set).

6 Max-Flow Min-Cut Theorem

Theorem Statement: In a flow network, the maximum amount of flow passing from the source to the sink is equal to the total weight of the edges in a minimum cut.

$$\text{Maximum Flow} = \text{Minimum Cut Capacity}$$

Definitions

- **Capacity of a cut:** The sum of the capacities of the edges in the cut oriented from set X to set Y .
- **Minimum cut:** The cut in the network with the smallest possible capacity.
- **Maximum flow:** The largest possible flow from source S to sink T .

7 Deterministic Min-Cut Algorithm

7.1 Stoer–Wagner Minimum Cut Algorithm

Let s and t be two vertices of graph G . Let $G/\{s, t\}$ be the graph obtained by merging s and t . A minimum cut of G can be obtained by taking the smaller of:

1. A minimum $s - t$ cut of G .
2. A minimum cut of $G/\{s, t\}$.

7.2 Pseudocode (Deterministic)

Algorithm 1 MinimumCutPhase(G, a)

```

 $A \leftarrow \{a\}$ 
while  $A \neq V$  do
    Add to  $A$  the most tightly connected vertex
end while
return The cut weight as the “cut of the phase”

```

Algorithm 2 MinimumCut(G)

```

while  $|V| \geq 1$  do
    Choose any  $a$  from  $V$ 
    MinimumCutPhase( $G, a$ )
    if cut-of-the-phase < current minimum cut then
        Store cut-of-the-phase as current minimum cut
    end if
    Shrink  $G$  by merging the two vertices added last
end while
return The minimum cut

```

8 Randomized Min-Cut Algorithm

8.1 Why Randomized Algorithm?

Randomized algorithms provide:

- A probabilistic guarantee of success.
- A more accurate estimate of the minimum cut with fewer iterations.
- Efficiency and ease of parallelization.

8.2 Karger's Algorithm Theorem

The algorithm outputs a min-cut set with probability at least:

$$\frac{2}{n(n-1)}$$

where n is the number of vertices.

9 Comparison: Deterministic vs Randomized

Feature	Deterministic (Stoer-Wagner)	Randomized (Karger)
Guarantee	Always guarantees an exact minimum cut.	Provides an approximate minimum cut with high probability.
Complexity	$O(V \cdot E + V^2 \log V)$	$O(V^2)$ (per run)

10 Python Simulation

- Open the Campuswire Post regarding Lecture 10.
- Download the `.ipynb` file pasted in the comments to run the simulation.

End of Lecture 10 Scribe